RV College of Engineering®
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DEPARTMENT OF PHYSICS

Academic year 2024-2025

(CIE I FOR FIRST SEM B.E. PROGRAMS: ME,AS,CH,IM)

Date	NOVEMEBER 04, 2024	Maximum Marks	50
Course Code	PY221IB	Duration	90min
Course name	**CLASSICAL PHYSICS**	Student ROLL.NO:	

Answer all questions

Sl. No.	QUESTIONS	M	BT	CO
1 a)	Starting from the differential equation of a damped oscillator, derive an expression for the effective frequency of a damped oscillator. Show the solution graphically.	7	1	1
1 b)	Calculate the time period of a spring and mass system, given the spring comprises of two equal springs of spring constant 10 N/m in series and the mass attached is 1 kg.	3	3	2
2 a)	Derive an expression for the amplitude and phase of a forced-damped oscillator. Calculate the specific values of these quantities when the oscillator is overdamped.	7	2	1
2 b)	Calculate the impedance and phase of a R-L series circuit where reactance of the inductor is 2Ω and resistance of resistor is 2Ω.	3	3	2
3 a)	What is resonance? Why does the resonant frequency shift to lower values as the damping term increases?	5	3	3
3 b)	Derive an expression for the time period of a torsional pendulum. How can we use this formula to determine the couple per unit twist.	5	2	1, 4
4 a)	With a neat diagram explain the Stress-Strain curve.	7	2	1
4 b)	Given the value of stress at the elastic limit is 100 N/m^2, calculate the area under the stress- strain curve till the elastic limit. Given the young's modulus of the material is 15×10^{10} N/m^2.	3	2	2
5 a)	With a neat labelled diagram prove that shearing stress acting on a fixed body equals Tensile stress along its diagonal.	7	3	1
5 b)	Calculate the Bulk Modulus of a body whose Poisson ratio = 0.2 and Youngs Modulus is 20×10^{10} N/m^2.	3	2, 3	2

M – Marks allotted, BT – Blooms Taxonomy, CO- Course Outcomes.

Course Outcomes	CO1	CO2	CO3	CO4
Max Marks	30	12	5	3

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Date	NOVEMEBER 04, 2024	Maximum Marks	50
Course Code	PY23HB	Duration	90min
Course name	**CLASSICAL PHYSICS**	Student ROLL.NO:	

Solution

Sl. No.	QUESTIONS	Soln
1 a)	Starting from the differential equation of a damped oscillator, derive an expression for the effective frequency of a damped oscillator. Show the solution graphically.	Diff eqn of damped osc – 1, trial soln – 1. Quadratic eqn in α and 2 roots of α – 1, underdamped soln -2, expression for effective frequency – 1 , Graphical soln for underdamped case -1 .
1 b)	Calculate the time period of a spring and mass system, given the spring comprises of two equal springs of spring constant 10 N/m in series and the mass is 1 kg.	2 equals springs in series , So eff spring constant = k/2 = 5 N/m. -1, $T = 2\pi(m/k)^{1/2} = 2\pi/2.236 = 2.81$ s - 2
2 a)	Derive an expression for the amplitude and phase of a forced damped oscillator. Calculate the specific values of these quantities in the limit of the forcing frequency being much larger than the natural frequency.	Diff eqn of forced, damped oscillator – 1 Trial soln and solving 2 eqns – 3, Expression for amplitude and phase – 2, Limiting value of A and φ when $\omega_0 >> \omega$ ---1
2 b)	Calculate the impedance and phase of a R-L series circuit where reactance of the inductor is 2Ω and resistance of resistor is 2Ω.	$Z = [X_L^2 + R^2]^{1/2} = (4+4)^{1/2} = 2.828\ \Omega$ --1 $\tan\varphi = X_L/R = 2/2 = 1$ ---1 .So $\varphi = 45^0$ ..1
3 a)	What is resonance? Why does the resonant frequency shift to lower values as the damping term increases?	When the driving frequency matches the natural frequency of an undamped oscillator, the amplitude of oscillation peaks to infinity. As **the damping term increases, the denominator cannot be zero. It has to be minimized which is possible only when the driving frequency is slightly less than the natural frequency.**
3 b)	Derive an expression for the time period of a torsional pendulum. How can we use this formula to determine the couple per unit twist experimentally?	Start from differential equation of torsional motion to arrive at expression for ω, T ....2 Attach a pendulum of known moment of inertia as pendulum. Measure its time period. Use formula above to calculate C.
4 a)	With a neat diagram explain the Stress-Strain curve.	Diagram – 2 , Elastic limit, permanent set, strain hardening, strain softening, ultimate stress – 1 mark each.
4 b)	Given the value of stress at the elastic limit is 100 N calculate the area under	Stress/Strain = Y . So Strain = Stress/Y ...1 Curve is linear till elastic limit. So area under

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	the curve till the elastic limit. Given Young's Modulus = 15×10^{10} N/m^2.	curve = (Stress x Strain)/2 = Stress^2/2Y …1 = $10000/(3 \times 10^{11}) = 3.33 \times 10^{-8}$ …..1
5 a)	With a neat labelled diagram prove that shearing stress acting on a fixed body equals Tensile stress along its diagonal.	Diagram – 2, Net force along 2 diagonals – 2, Area normal to Tensile stress …1, Area normal to Compressive stress … 1, Stress –1
5 b)	Calculate the Bulk Modulus of a body whose Poisson ratio = 0.2 and Young's Modulus is 20×10^{10} N/m^2.	$K = Y/[3(1 - 2\sigma)]$, $K = 2 \times 10^{11}/[3(1 - 0.4)] = 1.11 \times 10^{11}$ N/m^2

M – Marks allotted, BT – Blooms Taxonomy, CO- Course Outcomes.

Course Outcomes	CO1	CO2	CO3	CO4
Max Marks	30	12	5	3